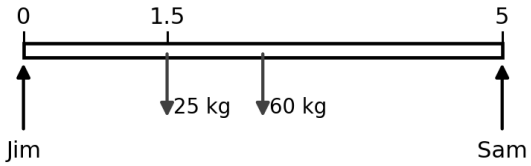
PHYS1408 Practice Exam 3-2

Chapters 12 (Statics) and 13 (Gravity) — 15 multiple-choice questions

$G = 6.67 \times 10^{-11}\ \frac{\text{N m}^2}{\text{kg}^2}$ $\quad M_{earth} = 5.98 \times 10^{24}\ \text{kg}$ $\quad R_{earth} = 6.37 \times 10^{6}\ \text{m}$
$g = 9.8\ \text{m/s}^2$

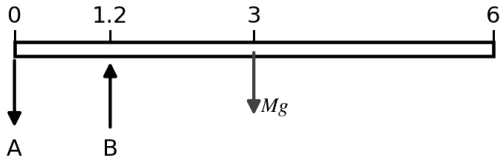
Instructions. Choose the single best answer (a–e). Answers are rounded; pick the nearest choice. For a **maximum** ("how far can he safely…"), pick the largest choice that still works; for a **minimum**, the smallest choice that still works. A calculator and your 3×5 notecard are allowed.

1. Jim and Sam each hold one end of a uniform 5 m, 60 kg sofa. A 25 kg child is placed on the sofa 1.5 m from Jim. How much force must Jim exert now?



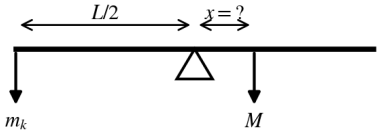
a) 47.5 N
b) 172 N
c) 294 N
d) 368 N
e) 466 N

2. A uniform diving board is 6 m long and has a mass of 400 kg. It rests on two supports: support A at the back end (x = 0) and support B 1.2 m from that end. Nothing is standing on the board. What is the magnitude of the force from support A?



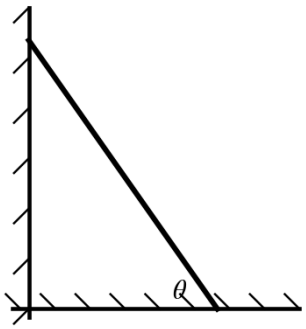
a) 3,920 N
b) 5,880 N
c) 9,800 N
d) 13,700 N
e) 15,700 N

3. A seesaw 4 m long pivots at its center. A 24 kg kid sits at one end. How far from the center must a 60 kg adult sit on the other side so that the seesaw balances? The seesaw itself has a mass of 30 kg.



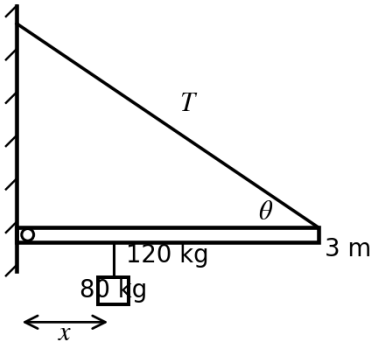
a) 0.533 m
b) 0.800 m
c) 1.20 m
d) 1.60 m
e) 2.80 m

4. A uniform ladder leans against a frictionless wall. The coefficient of static friction between the ladder and the floor is $\mu_s = 0.35$. What is the minimum angle θ between the ladder and the floor at which the ladder will not slide?



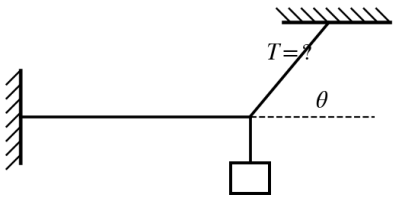
a) 19.3°
b) 35.0°
c) 55.0°
d) 70.7°
e) 80.3°

5. A uniform 3 m, 120 kg beam sticks straight out from a wall on a hinge. A cable runs from the far end of the beam back to the wall above the hinge, making a 35° angle with the beam. The cable can hold at most $T_{max} = 2{,}000$ N. A 80 kg mass hangs from the beam a distance x from the wall. What is the largest x the cable can support?



a) 0.390 m
b) 0.546 m
c) 0.764 m
d) 1.07 m
e) 2.14 m

6. A 12 kg light hangs from a knot. One cord runs from the knot to the ceiling, making a 50° angle with the ceiling. A second cord is exactly horizontal and is anchored in the wall. What is the tension in the cord connected to the ceiling?

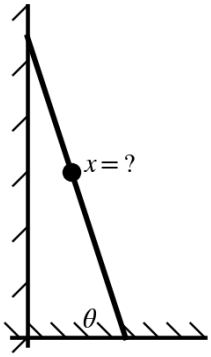


a) 70.5 N
b) 98.7 N
c) 118 N
d) 154 N
e) 183 N

7. A 150 kg load hangs from a circular wire 2 m long with a diameter of 0.001 m and a Young's modulus of 7×10^{10} N/m^2. By how much does the wire stretch?

a) 0.00546 m
b) 0.0134 m
c) 0.0267 m
d) 0.0420 m
e) 0.0535 m

8. A 140 kg man (with the load he carries) climbs a uniform 9 m, 15 kg ladder that leans against a frictionless wall at 72° to the floor. The coefficient of static friction between the ladder and the floor is 0.3. How far up the ladder can he safely climb?



a) 8.5 m
b) 9.0 m
c) 9.5 m
d) 10.0 m
e) 10.5 m

9. What is the acceleration due to gravity at a height of 1{,}200 km above the Earth's surface?

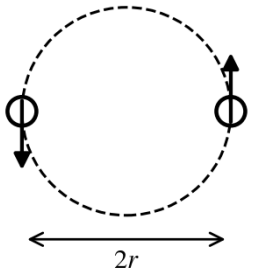
a) 6.96 m/s^2
b) 8.25 m/s^2
c) 9.83 m/s^2
d) 277 m/s^2
e) 388 m/s^2

10. A satellite is to circle the Earth once every 12 hours. At what altitude above the Earth's surface must it orbit?

a) 10,400 km
b) 20,200 km
c) 26,600 km
d) 84,300 km
e) 1.37×10^5 km

11. Two stars, each with the mass of the Sun (2×10^{30} kg), circle their common center of mass. Each star moves on a circle of radius 3×10^{10} m. What is the period?

a) 32.7 days
b) 46.3 days
c) 65.4 days
d) 92.5 days
e) 185 days



12. A rocket is launched straight up from the Earth's surface at 2,000 m/s. How high above the surface does it rise? (Ignore air resistance.)

a) 107 km
b) 150 km
c) 210 km
d) 435 km
e) 6,580 km

13. A black hole has a mass of 4×10^{36} kg. If its escape speed equals the speed of light ($c = 3 \times 10^8$ m/s), what is the radius of its event horizon?

a) 55,000 m
b) 77,000 m
c) 2.96×10^9 m
d) 5.93×10^9 m
e) 1.19×10^{10} m

14. An asteroid orbits the Sun at 2.8 AU (1 AU $= 1.5 \times 10^{11}$ m, $M_{sun} = 2 \times 10^{30}$ kg). Treating the orbit as a circle, what is its orbital speed?

a) 17.8 km/s
b) 25.2 km/s
c) 29.8 km/s
d) 41.7 km/s
e) 58.4 km/s

15. The Earth and the moon (mass 1.2×10^{23} kg) are 500,000 km apart, center to center. How far from the center of the Earth toward the moon, as a percentage of the Earth–moon distance, is the point where the gravitational pull toward the Earth equals the pull toward the moon?

a) 12.4%
b) 50.0%
c) 78.6%
d) 87.6%
e) 98.0%